

Question Number	Answer	Additional guidance	Mark
3(a)	An answer that makes reference to three of the following: - cell wall with secondary thickening {is thicker / has more layers} (1) - cell wall with secondary thickening contains lignin (1) - cell wall with {secondary thickening / lignin} is {stronger / more supported / less flexible / has pits} (1) - cell wall with {secondary thickening / lignin} is {waterproof / impermeable to water} (1)	accept converse for all mp accept diagram 2 for cell wall with {secondary thickening / lignin} Mark as a whole as some statements may have more than 1 mp in them accept cell wall with secondary thickening has greater diffusion distance accept secondary thickening layer is thinner than original cell wall accept stability ignore cellulose giving these properties ignore harder ignore insoluble	(3)

Question Number	Answer	Mark
3(b)(i)	The only correct answer is B one <i>A is not correct because cellulose molecules form microfibrils, molecules do not contain 1,6 glycosidic bonds and do not contain α-glucose</i> <i>C is not correct because cellulose molecules form microfibrils, molecules do not contain 1,6 glycosidic bonds and do not contain α-glucose</i> <i>D is not correct because cellulose molecules form microfibrils, molecules do not contain 1,6 glycosidic bonds and do not contain α-glucose</i>	(1)

Question Number	Answer	Additional guidance	Mark
3(b)(ii)	An explanation that makes reference to the following points: • (secondary thickening involves addition of) lignin (1) • which results in (increased) strength (1) IMP	ignore cellulose accept lignified / lignification accept more rigid/ rigidity / waterproof / impermeable to water ignore support / harder	(2)

Question Number	Answer	Additional guidance	Mark
3(c)	A description that makes reference to the following points: - allows {movement of named substances / communication} (1) - between (adjoining) cells / to other tissues / out of xylem / into xylem (1) IMP	e.g. water, inorganic ions, mineral ions ignore nutrients accept {lateral / sideways} movement ignore movement {up/through} xylem accept connects xylem vessels lateral movement of water = 2 marks	(2)